KALAI NITHI GUHAN M

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Objective:

Motivated Computer Science student seeking an entry-level opportunity to apply my technical skills in a dynamic and growth-oriented organization. Passionate about problem-solving, technology, and continuous learning, with hands-on experience in developing practical applications and exploring emerging technologies.

Area of Interest:

- **Machine Learning & AI** (Model Development, AI Agents, Data Processing)
- **Computer Vision** (OpenCV, MediaPipe, YOLO, Image Processing)
- **Natural Language Processing** (LLMs, RAG)

Education:

- **Bachelor of Engineering in Computer Science and Engineering (Artificial Intelligence and Machine Learning)**
  Dr. Mahalingam College of Engineering and Technology, Pollachi, Coimbatore, CGPA: 8.63*. **2023 to 2027**
- **HSC** | Srinivasa Vidhyalaya Matriculation Higher Secondary School, Udumalpet, 89.3%. **2022 to 2023**
- **SSLC** | Srinivasa Vidhyalaya Matriculation Higher Secondary School, Udumalpet. **2020 to 2021**

Skills:

Technical Skills:

- **Programming Languages:** Java, Python, C
- **Frameworks:** TensorFlow, PyTorch, MediaPipe, Hugging Face Transformers
- **Cloud & Database:** Firebase, Vercel, Render, MySQL
- **Tools:** Git, GitHub, VS Code, Ollama
- **Web Development:** Next.js, Node.js, Tailwind CSS

Soft Skills:

- **Problem-Solving:** Strong analytical thinking and algorithmic coding skills
- **Quick Learner:** Quickly adapts to new technologies, tools, and working environments.
- **Team Player:** Able to collaborate effectively in team environments

Projects:

Autonomous AI Agent System **December to January 2026**

- Implemented an autonomous AI agent powered by Meta's Llama 3.1 (8B) via Ollama for local inference and reasoning.
- Integrated real-time web scraping using BeautifulSoup to retrieve and process dynamic online data.
- Architected a Next.js interface supporting real-time interaction and deployed the system for scalable user access.

Contextual Document QA System (RAG-LLM) **January to February 2026**

- Developed a Retrieval-Augmented Generation (RAG) application integrating LLMs, vector databases, and semantic search for intelligent document question answering.
- Built a FastAPI backend and Next.js frontend to enable PDF upload, retrieval, and context-aware response generation.
- Integrated Pinecone, Sentence Transformers, and OpenRouter for efficient document indexing and real-time AI-powered responses.

Exercise Counter using OpenCV **October to November 2025**

- Built a real-time computer vision exercise tracking system using OpenCV and MediaPipe achieving 90% pose detection accuracy.
- Configured real-time webcam pose detection to count push-ups, squats, and bicep curls.
- Achieved an 80% functional prototype with reliable motion analysis.

Internships:

Boredom LLP Technology, Bangalore. **May to June 2025**

- Developed an AI-based fertilizer recommendation system that generates intelligent fertilizer suggestions from user inputs.
- Designed 3+ responsive web pages using HTML, CSS, and JavaScript for seamless cross-device performance.
- Built interactive front-end components using React to enhance user experience and integrated Firebase for real-time data handling.
- Worked with Next.js architecture including project structure, routing mechanisms, and basic API integration.

Certifications:

- Completed Natural Language Processing (NLP) Certification from PrepInsta Prime.

Extracurricular Activities:

- Solved 650+ Data Structures and Algorithms problems on LeetCode, improving problem-solving and algorithmic thinking skills.